

Example 8: Let $A = \{a, e, i, o, u\}$ (where the universal set is the set of letters of the English alphabet). Then
 $\bar{A} = \{b, c, d, f, g, h, j, k, l, m, n, p, q, r, s, t, v, w, x, y, z\}$ $\square$

2.2.2 Set Identities

TABLE 1 Set Identities	
Identity	Name
$A \cap U = A$ $A \cup \emptyset = A$	Identity Laws
$A \cup U = U$ $A \cap \emptyset = \emptyset$	Domination Laws
$A \cap A = A$ $A \cup A = A$	Idempotent Laws
$\overline{(\bar{A})} = A$	Complementation Law
$A \cup B = B \cup A$ $A \cap B = B \cap A$	Commutative Laws
$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$	Associative Laws
$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$	Distributive Laws
$\overline{A \cap B} = \bar{A} \cup \bar{B}$ $\overline{A \cup B} = \bar{A} \cap \bar{B}$	De Morgan's Laws
$A \cup (A \cap B) = A$ $A \cap (A \cup B) = A$	Absorption Laws
$A \cup \bar{A} = U$ $A \cap \bar{A} = \emptyset$	Complement Laws

Example 10: Prove that $\overline{A \cap B} = \bar{A} \cup \bar{B}$

Solution: We will prove that the two sets $\overline{A \cap B}$ and $\bar{A} \cup \bar{B}$ are equal by showing that each set is a subset of the other.

• Let $x \in \overline{A \cap B}$. By Defn. 5, $x \notin A \cap B$. By Defn. 2, $x \notin A \wedge x \notin B$. By Defn. of AND, $x \notin A$ and $x \notin B$. By Defn. 5, $x \in \bar{A}$. By Defn. 5, $x \in \bar{B}$. By Defn. 1, $x \in \bar{A} \cup \bar{B}$. $\therefore$ By Defn. 3 of Chapter 2.1, $\boxed{\overline{A \cap B} \subseteq \bar{A} \cup \bar{B}}$ since x is an arbitrary element of $\overline{A \cap B}$. (Wrong Proof)